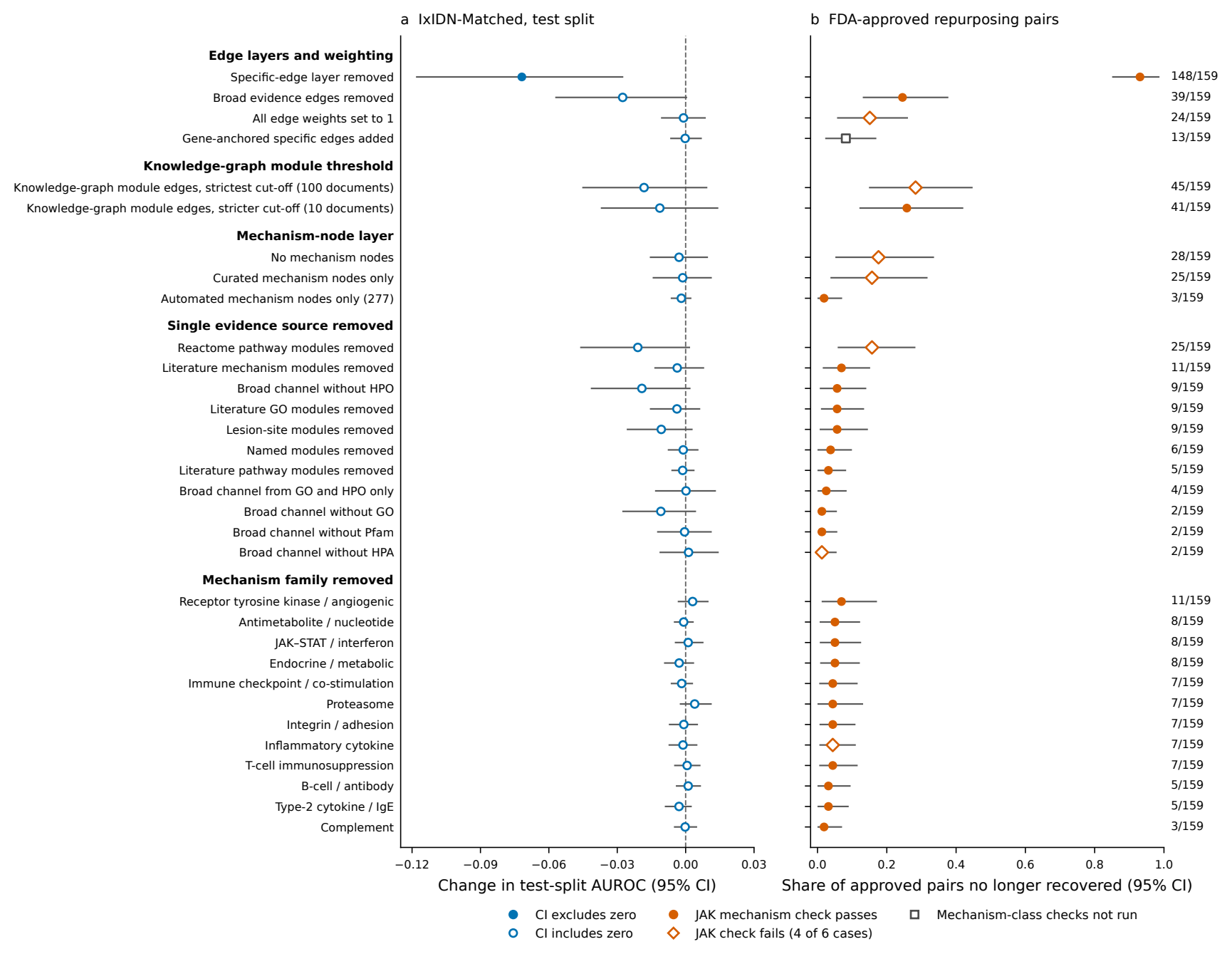
a IxIDN-Matched, test split
b FDA-approved repurposing pairs
Edge layers and weighting
Specific-edge layer removed
Broad evidence edges removed
All edge weights set to 1
Gene-anchored specific edges added
Knowledge-graph module threshold
Knowledge-graph module edges, strictest cut-off (100 documents)
Knowledge-graph module edges, stricter cut-off (10 documents)
Mechanism-node layer
No mechanism nodes
Curated mechanism nodes only
Automated mechanism nodes only (277)
Single evidence source removed
Reactome pathway modules removed
Literature mechanism modules removed
Broad channel without HPO
Literature GO modules removed
Lesion-site modules removed
Named modules removed
Literature pathway modules removed
Broad channel from GO and HPO only
Broad channel without GO
Broad channel without Pfam
Broad channel without HPA
Mechanism family removed
Receptor tyrosine kinase / angiogenic
Antimetabolite / nucleotide
JAK–STAT / interferon
Endocrine / metabolic
Immune checkpoint / co-stimulation
Proteasome
Integrin / adhesion
Inflammatory cytokine
T-cell immunosuppression
B-cell / antibody
Type-2 cytokine / IgE
Complement
148/159
39/159
24/159
13/159
45/159
41/159
28/159
25/159
3/159
25/159
11/159
9/159
9/159
9/159
6/159
5/159
4/159
2/159
2/159
2/159
11/159
8/159
8/159
8/159
7/159
7/159
7/159
7/159
7/159
5/159
5/159
3/159
−0.12
−0.09
−0.06
−0.03
0.00
0.03
0.0
0.2
0.4
0.6
0.8
1.0
Change in test-split AUROC (95% CI)
Share of approved pairs no longer recovered (95% CI)
CI excludes zero
CI includes zero
JAK mechanism check passes
JAK check fails (4 of 6 cases)
Mechanism-class checks not run